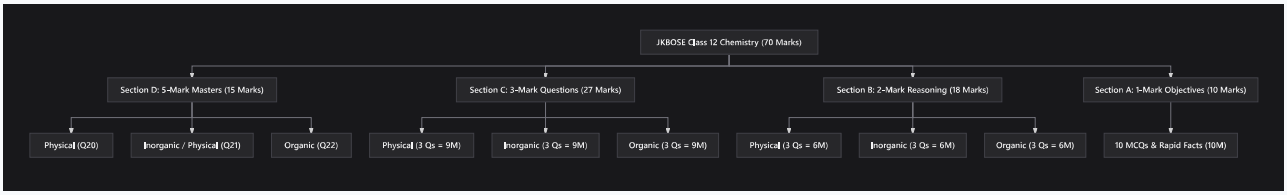
JKBOSE Class 12 Chemistry: Ultimate Categorized Blueprint (70/70 Hard-Paper Shield)

> **Board**: Jammu & Kashmir Board of School Education (JKBOSE)
> **Theory Marks**: 70 Marks | **Practicals**: 30 Marks | **Total**: 100 Marks
> **Structure**: Categorized strictly into **5-Mark Questions**, **3-Mark Questions**, **2-Mark Questions**, and **1-Mark Objectives**, subdivided into **Physical**, **Inorganic**, and **Organic Chemistry**.

Blueprint Overview & Marks Distribution



PART 1: THE 5-MARK LONG ANSWER MASTERS (SECTION D — 15 MARKS)

In JKBOSE, Section D contains 3 questions with 100% internal choice. Every long question consists of a main derivation/mechanism (3M) and an attached numerical/sub-question (2M).

1.1 Physical Chemistry 5-Mark Masters

Question 1 [Kinetics & Rate Laws]

(a) Derive the integrated rate expression for a First-Order Reaction. Prove that its half-life period ($t_{1/2}$ t1/2) is completely independent of the initial concentration of reactants. (3 Marks)

- **Step 1 (Differential equation)**: For a general reaction $R \rightarrow P$R → P:
$$-\frac{d[R]}{dt} = k[R]^1 \implies \frac{d[R]}{[R]} = -kdt$$
−dtd[R] = k[R]1 ⟹ [R]d[R] = −k dt
- **Step 2 (Integration)**:
$$\int \frac{d[R]}{[R]} = -k \int dt \implies \ln[R] = -kt + I \quad \text{— (Eq 1)}$$
∫ [R]d[R] = −k ∫ dt ⟹ ln[R] = −kt + I — (Eq 1)
- **Step 3 (Integration constant)**: At $t = 0$t = 0, $[R] = [R]_0 \implies \ln[R]_0 = -k(0) + I \implies I = \ln[R]_0$ [R] = [R]0 ⟹ ln[R]0 = −k(0) + I ⟹ I = ln[R]0.
Substituting II back: $\ln[R] = -kt + \ln[R]_0 \implies kt = \ln\frac{[R]_0}{[R]}$ln[R] = −kt + ln[R]0 ⟹ kt = ln [R][R]0.
- **Step 4 (Base-10 formula)**:
$$k = \frac{2.303}{t}\log_{10}\frac{[R]_0}{[R]} \quad \text{or} \quad t = \frac{2.303}{k}\log_{10}\frac{[R]_0}{[R]}$$
k = t2.303 log10 [R][R]0 or t = k2.303 log10 [R][R]0
- **Step 5 (Half-life proof)**: At $t = t_{1/2}$ t = t1/2, $[R] = \frac{[R]_0}{2}$[R] = 2[R]0:
$$t_{1/2} = \frac{2.303}{k}\log_{10}\left(\frac{[R]_0}{[R]_0/2}\right) = \frac{2.303 \times \log_{10}(2)}{k} = \frac{0.693}{k}$$
t1/2 = k2.303 log10 ([R]0/2[R]0) = k2.303 × log10(2) = k0.693*Conclusion*: The expression contains no concentration term $[R]_0$ [R]0, proving $t_{1/2}$ t1/2 is completely independent of initial concentration.

(b) A first-order reaction has a rate constant $k = 1.15 \times 10^{-3}\ \text{s}^{-1}$ k = 1.15 × 10−3 s−1. How long will $5\ \text{g}$5 g of this reactant take to reduce to $3\ \text{g}$3 g? (2 Marks)

- $[R]_0 = 5\ \text{g}$[R]0 = 5 g, $[R] = 3\ \text{g}$[R] = 3 g, $k = 1.15 \times 10^{-3}\ \text{s}^{-1}$ k = 1.15 × 10−3 s−1.
- $$t = \frac{2.303}{k}\log_{10}\frac{[R]_0}{[R]} = \frac{2.303}{1.15 \times 10^{-3}}\log_{10}\left(\frac{5}{3}\right) = \frac{2.303}{1.15 \times 10^{-3}} \times (\log 5 - \log 3)$$
t = k2.303 log10 [R][R]0 = 1.15 × 10−32.303 log10 (35) = 1.15 × 10−32.303 × (log 5 − log 3)
- $t = \frac{2.303}{1.15 \times 10^{-3}} \times (0.6990 - 0.4771) = 2002.6 \times 0.2219 = 444.38\ \text{seconds}$t = 1.15×10−32.303 × (0.6990 − 0.4771) = 2002.6 × 0.2219 = 444.38 seconds.

Question 2 [Electrochemistry & Conductivity]

(a) State Kohlrausch's Law of Independent Migration of Ions. Explain its two main applications with expressions. (3 Marks)

- **Statement**: "*At infinite dilution, the limiting molar conductivity (Λ_m° Λm∘) of an electrolyte is equal to the sum of the individual contributions of its constituent cation and anion.*"
$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$
Λm∘ = ν+λ+∘ + ν−λ−∘

- **Application 1 (Weak Electrolytes)**: Calculate Λ_m° Λm∘ for weak electrolytes like acetic acid using strong electrolytes:
$$\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl)$$
Λm∘(CH3COOH) = Λm∘(CH3COONa) + Λm∘(HCl) − Λm∘(NaCl)
- **Application 2 (Degree of Dissociation & K_a Ka)**:
$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}, \qquad K_a = \frac{c\alpha^2}{1-\alpha}$$
α = Λm∘Λm, Ka = 1 − αcα2

(b) Calculate the EMF and ΔG° ΔG∘ for the cell at $298\ K$298 K: (2 Marks)
$$Mg(s) \mid Mg^{2+}(0.001\ M) \parallel Cu^{2+}(0.0001\ M) \mid Cu(s)$$
Mg(s) | Mg2+(0.001 M) ‖ Cu2+(0.0001 M) | Cu(s)
(Given: $E^\circ_{Mg^{2+}/Mg} = -2.37\ V$EMg2+/Mg∘ = −2.37 V, $E^\circ_{Cu^{2+}/Cu} = +0.34\ V$ECu2+/Cu∘ = +0.34 V, $F = 96500\ C \cdot mol^{-1}$ F = 96500 C ⋅mol−1)

- $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-2.37) = +2.71\ V$Ecell∘ = Ecathode∘ − Eanode∘ = 0.34 − (−2.37) = +2.71 V, $n = 2$n = 2.
- $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log_{10}\frac{[Mg^{2+}]}{[Cu^{2+}]} = 2.71 - \frac{0.0591}{2}\log_{10}(\frac{10^{-3}}{10^{-4}}) = 2.71 - 0.02955(1) = 2.68\ V$Ecell = Ecell∘ − n0.0591 log10 [Cu2+][Mg2+] = 2.71 − 20.0591 log10 (10−410−3) = 2.71 − 0.02955(1) = 2.68 V.
- $\Delta G^\circ = -nFE^\circ_{cell} = -2 \times 96500 \times 2.71 = -523030\ J \cdot mol^{-1} = -523.03\ kJ \cdot mol^{-1}$ ΔG∘ = −nF Ecell∘ = −2 × 96500 × 2.71 = −523030 J ⋅ mol−1 = −523.03 kJ ⋅ mol−1.

Question 3 [Solutions & Colligative Properties]

(a) What are Colligative Properties? Derive the relationship between Depression in Freezing Point (ΔT_f ΔTf) and molar mass of solute (M_2 M2). (3 Marks)

- **Colligative Properties**: Properties that depend solely on the total number of solute particles in solution, independent of their chemical nature.
- **Derivation**:
$\Delta T_f = T_f^\circ - T_f \propto m \implies \Delta T_f = K_f \cdot m$ΔTf = Tf∘ − Tf ∝ m ⟹ ΔTf = Kf ⋅ m.
Molality $m = \frac{w_2 \times 1000}{M_2 \times w_1}$m = M2×w1w2×1000 ($w_2$ w2 = solute mass, w_1 w1 = solvent mass in grams).
$$\Delta T_f = K_f \times \frac{w_2 \times 1000}{M_2 \times w_1} \implies M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1}$$
ΔTf = Kf × M2 × w1w2 × 1000 ⟹ M2 = ΔTf × w1Kf × w2 × 1000

(b) What is Van 't Hoff factor (ii)? Write its relation with degree of dissociation (αα) and calculate ii for complete dissociation of $K_4[Fe(CN)_6]$K4[Fe(CN)6]. (2 Marks)

- $i = \frac{\text{Normal Molar Mass}}{\text{Abnormal Molar Mass}} = \frac{\text{Observed Colligative Property}}{\text{Calculated Colligative Property}}$i = Abnormal Molar MassNormal Molar Mass = Calculated Colligative PropertyObserved Colligative Property.
- $\alpha = \frac{i-1}{n-1}$α = n−1i−1. For complete dissociation ($\alpha = 1$α = 1): $i = n$i = n.

- $K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-} \implies n = 4 + 1 = 5 \implies i =$ 5K4[Fe(CN)6] → 4K+ + [Fe(CN)6]4− ⟹ n = 4 + 1 = 5 ⟹ i = 5.

1.2 Inorganic Chemistry 5-Mark Masters

Question 4 [Coordination Chemistry & d-Block]

(a) Using Valence Bond Theory (VBT), predict the hybridization, geometry, and magnetic behavior of: (3 Marks)

1. $[Fe(CN)_6]^{3-}$ **[Fe(CN)6]3−** (Fe = 26Fe = 26): Fe^{3+} Fe3+ is $3d^5$ 3d5. CN^- CN− is a strong field ligand, forcing pairing $\implies d^2sp^3 \implies$ d2sp3 hybridization, **Octahedral** geometry, **Paramagnetic** (11 unpaired electron, low-spin inner orbital complex).
2. $[Ni(CN)_4]^{2-}$ **[Ni(CN)4]2−** (Ni = 28Ni = 28): Ni^{2+} Ni2+ is $3d^8$ 3d8. CN^- CN− forces pairing into four $3d$3d orbitals, leaving one empty $3d$3d $\implies dsp^2 \implies$ dsp2 hybridization, **Square Planar** geometry, **Diamagnetic** (00 unpaired electrons).
3. $[NiCl_4]^{2-}$ **[NiCl4]2−**: Cl^- Cl− is a weak field ligand, cannot pair electrons $\implies sp^3 \implies$ sp3 hybridization, **Tetrahedral** geometry, **Paramagnetic** (22 unpaired electrons).

(b) What is Lanthanoid Contraction? State its primary cause and two major consequences on transition elements. (2 Marks)

- **Definition**: Gradual decrease in atomic and ionic radii across lanthanoids ($La^{3+} \rightarrow Lu^{3+}$ La3+ → Lu3+).
- **Cause**: Poor, diffuse shielding of nuclear charge by inner $4f$4f electrons, increasing effective nuclear charge (Z_{eff} Zeff).
- **Consequences**: (1) Identical atomic sizes for $4d$4d and $5d$5d pairs (e.g., $Zr \approx Hf$ Zr ≈ Hf). (2) Basic strength of hydroxides decreases from $La(OH)_3$ La(OH)3 to $Lu(OH)_3$ Lu(OH)3.

1.3 Organic Chemistry 5-Mark Masters

Question 5 [Aldehydes, Ketones & Carboxylic Acids]

(a) Write balanced chemical equations and conditions for: (3 Marks)

1. **Aldol Condensation**:

$$2CH_3CHO \xrightarrow{\text{dil. NaOH}} CH_3 - CH(OH) - CH_2 - CHO \xrightarrow{\Delta,\ -H_2O} CH_3 - CH = CH - CHO \text{ (But-2-enal)}$$

2CH3CHO dil. NaOH

CH3 – CH(OH) – CH2 – CHO Δ,−H2O

CH3 – CH = CH – CHO (But-2-enal)

2. **Cannizzaro Reaction**:
$2C_6H_5CHO + \text{conc. } KOH \xrightarrow{\Delta} C_6H_5COOK \text{ (Potassium benzoate)} + C_6H_5CH_2OH \text{ (Benzyl alcohol)}$
2C6H5CHO + conc. KOH Δ

C6H5COOK (Potassium benzoate) + C6H5CH2OH (Benzyl alcohol)

3. **Hell-Volhard-Zelinsky (HVZ) Reaction**:
$CH_3COOH + Cl_2 \xrightarrow{\text{Red P}} CH_2Cl\text{-}COOH \ (\alpha\text{-chloroacetic acid}) + HCl$
CH3COOH + Cl2 Red P

CH2Cl-COOH (α-chloroacetic acid) + HCl

(b) Give simple chemical tests to distinguish between: (2 Marks)

1. **Ethanal and Propanal**: **Iodoform Test**. Ethanal contains the $CH_3C=O$CH3C = O group and gives a **yellow precipitate of** CHI_3 **CHI3** with $I_2 + NaOH$I2 + NaOH; Propanal does not.
2. **Benzoic Acid and Phenol**: $NaHCO_3$ **NaHCO3 Test**. Benzoic acid produces **brisk effervescence of** CO_2 **CO2 gas**; Phenol does not react with $NaHCO_3$ NaHCO3.

Question 6 [Alcohols, Phenols & Amines Mechanisms]

(a) Write the stepwise mechanism for the Acid-Catalyzed Dehydration of Ethanol to Ethene at $443\ K$**443 K. (3 Marks)**

- **Step 1 (Protonation of alcohol)**:
$CH_3CH_2OH + H^+ \rightleftharpoons CH_3CH_2OH_2^+$ (Ethyl oxonium ion)
CH3CH2OH + H+ ⇌ CH3CH2OH2+ (Ethyl oxonium ion)
- **Step 2 (Formation of Carbocation — Slow, RDS)**:
$CH_3CH_2OH_2^+ \xrightarrow{\text{slow}} CH_3CH_2^+ + H_2O$
CH3CH2OH2+ slow

CH3CH2+ + H2O

- **Step 3 (Elimination of Proton)**:
$CH_3CH_2^+ \rightarrow CH_2=CH_2 + H^+$
CH3CH2+ → CH2 = CH2 + H+
(Hard paper shock-absorber: At $413\ K$*413 K, excess ethanol reacts via* S_N2*SN2 attack to yield Diethyl Ether* $CH_3CH_2OCH_2CH_3$ *CH3CH2OCH2CH3).*

(b) Write equations for: (2 Marks)

1. **Kolbe's Reaction**:
   $C_6H_5ONa + CO_2 \xrightarrow{400K, 4-7\ atm} \text{Sodium salicylate} \xrightarrow{H^+} \text{Salicylic acid}$C6H5ONa + CO2 400 K, 4−7 atm

   Sodium salicylate H+

   Salicylic acid
2. **Reimer-Tiemann Reaction**:
   $C_6H_5OH + CHCl_3 + 3NaOH \xrightarrow{340K} \text{Salicylaldehyde} + 3NaCl + 2H_2O$C6H5OH + CHCl3 + 3NaOH 340 K

   Salicylaldehyde + 3NaCl + 2H2O

PART 2: THE 3-MARK QUESTIONS (SECTION C — 27 MARKS)

2.1 Physical Chemistry 3-Markers

- **Q7 [Solutions] Differentiate between Ideal and Non-Ideal Solutions showing Positive and Negative Deviations with two points and one example each.**
  - *Ideal*: $\Delta H_{mix} = 0$ΔHmix = 0, $\Delta V_{mix} = 0$ΔVmix = 0, obeys Raoult's law over entire range (e.g., Benzene + Toluene).
  - *Positive Deviation*: $A-B$A − B forces are weaker than $A-A$A − A and $B-B$B − B. $\Delta H_{mix} > 0$ ΔHmix > 0, $\Delta V_{mix} > 0$ΔVmix > 0, forms **minimum boiling azeotropes** (e.g., Ethanol + Acetone).
  - *Negative Deviation*: $A-B$A − B forces are stronger than $A-A$A − A and $B-B$B − B. $\Delta H_{mix} < 0$ΔHmix < 0, $\Delta V_{mix} < 0$ΔVmix < 0, forms **maximum boiling azeotropes** (e.g., Acetone + Chloroform).
- **Q8 [Kinetics] Show that for a first-order reaction, time required for 99.9% completion is 10 times the half-life ($t_{1/2}$ t1/2).**
  - $t_{99.9\%} = \frac{2.303}{k}\log_{10}\left(\frac{100}{100-99.9}\right) = \frac{2.303}{k}\log_{10}(10^3) = \frac{2.303\times 3}{k} = \frac{6.909}{k}$t99.9% = k2.303log10 (100−99.9100) = k2.303log10(103) = k2.303×3 = k6.909.
  - $t_{1/2} = \frac{0.693}{k}$t1/2 = k0.693.
  - Ratio: $\frac{t_{99.9\%}}{t_{1/2}} = \frac{6.909/k}{0.693/k} = 9.97 \approx 10$t1/2t99.9% = 0.693/k6.909/k = 9.97 ≈ 10. Hence, $t_{99.9\%} = 10 \times t_{1/2}$ t99.9% = 10 × t1/2. (Proved).

- **Q9 [Kinetics] Write the Arrhenius Equation. How does a catalyst increase the rate of reaction?**
    - Formula: $k = Ae^{-E_a/RT}$ k = Ae−Ea/RT or $\log_{10}\frac{k_2}{k_1} = \frac{E_a}{2.303R}[\frac{T_2 - T_1}{T_1 T_2}]$log10 k1k2 = 2.303REa [T1T2 T2 −T1].
    - A catalyst provides an alternative reaction pathway with **lower activation energy (E_a Ea)**, increasing the fraction of effective collisions.

2.2 Inorganic Chemistry 3-Markers

- **Q10 [d-Block] Give concise scientific reasons for:**
    1. *Transition metals act as good catalysts*: Due to variable oxidation states and large surface area to adsorb reactants.
    2. *Transition metal compounds are colored in aqueous solution*: Due to $d - d$**d – d electron transitions** in split dd-orbitals.
    3. Cr^{2+} *Cr2+ is reducing while* Mn^{3+} *Mn3+ is oxidizing*: Cr^{3+} Cr3+ has stable half-filled t_{2g}^3 t2g3 configuration, while Mn^{2+} Mn2+ has stable half-filled d^5 d5 configuration.
- **Q11 [Coordination] Write the IUPAC names of:**
    1. $[\mathrm{Co(NH_3)_5Cl}]\mathrm{Cl_2}$ [Co(NH3)5Cl]Cl2 → → **Pentaamminechloridocobalt(III) chloride**
    2. $\mathrm{K_3[Fe(CN)_6]}$K3[Fe(CN)6] → → **Potassium hexacyanidoferrate(III)**
    3. $[\mathrm{Pt(NH_3)_2Cl(NO_2)}]$[Pt(NH3)2Cl(NO2)] → → **Diamminechloridonitrito-N-platinum(II)**
- **Q12 [Coordination] Explain Crystal Field Splitting in an Octahedral complex with an energy diagram. State when a high-spin complex is formed.**
    - Octahedral field splits degenerate dd-orbitals into higher energy e_g **eg** ($+0.6\Delta_o$ +0.6Δo) and lower energy t_{2g} **t2g** ($-0.4\Delta_o$ −0.4Δo).
    - High-spin complex forms when $\Delta_o < P$**Δo < P** (Pairing energy), with weak field ligands ($\mathrm{F^-}, \mathrm{Cl^-}$ F−, Cl−).

2.3 Organic Chemistry 3-Markers

- **Q13 [Haloalkanes] Compare S_N1SN1 and S_N2SN2 mechanisms on:**
    1. *Number of steps*: S_N1SN1 is 2 steps (carbocation intermediate); S_N2SN2 is 1 step (transition state).

2. *Reactivity order*: S_N1SN1 is $3^\circ > 2^\circ > 1^\circ$ 3∘ > 2∘ > 1∘; S_N2SN2 is $1^\circ > 2^\circ > 3^\circ$ 1∘ > 2∘ > 3∘ (least steric hindrance).
3. *Stereochemical outcome*: S_N1SN1 gives **Racemization**; S_N2SN2 gives **Inversion of configuration (Walden Inversion)**.

- **Q14 [Amines] Write balanced chemical equations for:**
  1. *Hoffmann Bromamide Degradation*:
     $CH_3CONH_2 + Br_2 + 4NaOH \rightarrow CH_3NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$ CH3CONH2 + Br2 + 4NaOH → CH3NH2 + Na2CO3 + 2NaBr + 2H2O
  2. *Carbylamine Reaction*:
     $C_2H_5NH_2 + CHCl_3 + 3KOH(alc) \xrightarrow{\Delta} C_2H_5NC\ (Foul\ smell) + 3KCl + 3H_2O$ C2H5NH2 + CHCl3 + 3KOH(alc) Δ

     C2H5NC (Foul smell) + 3KCl + 3H2O
  3. *Diazotization & Sandmeyer*: $C_6H_5NH_2 \xrightarrow{NaNO_2/HCl,\,0-5^\circ C} C_6H_5N_2^+Cl^- \xrightarrow{CuCl/HCl} C_6H_5Cl + N_2$ C6H5NH2 NaNO2 /HCl,0−5∘C

     C6H5N2+Cl− CuCl/HCl

     C6H5Cl + N2
- **Q15 [Biomolecules] Answer the following:**
  1. *What is Denaturation of Proteins?*: Disruption of native conformation by heat or pH. **Secondary and tertiary structures are destroyed; primary structure remains intact**.
  2. *What is Invert Sugar?*: Equimolar mixture of glucose and fructose formed by hydrolysis of sucrose, rotating light to the **laevo** direction.
  3. *Name the linkage joining amino acids in proteins*: **Peptide linkage (**$-CO-NH-$ **−CO – NH−)**.

PART 3: THE 2-MARK QUESTIONS (SECTION B — 18 MARKS)

3.1 Physical Chemistry 2-Markers

- **Q16 [Solutions] State Henry's Law. Why are aquatic species more comfortable in cold water than in warm water?**
  Statement: $p = K_H \cdot x$p = KH ⋅ x. K_H KH increases with temperature, which decreases the solubility of oxygen in water. Cold water has lower K_H KH and higher dissolved oxygen.
- **Q17 [Solutions] Define Osmotic Pressure (ΠΠ). Why is it preferred to find the molar mass of polymers?**
  Answer: Pressure applied on solution to prevent osmosis ($\Pi = CRT$Π = CRT). Preferred because measurements are carried out at room temperature and produce large, easily measurable values for high molecular mass solutes.
- **Q18 [Electrochemistry] Why does conductivity (κκ) decrease on dilution, while molar conductivity (Λ_m Λm) increases?**
  Answer: Conductivity (κκ) decreases because the **number of ions per unit volume decreases**. Molar conductivity ($\Lambda_m = \frac{\kappa \times 1000}{M}$Λm = Mκ×1000) increases because the increase in volume containing $1\ \text{mole}$1 mole of electrolyte compensates for the drop in κκ.
- **Q19 [Kinetics] Differentiate between Order and Molecularity of a reaction.**
  Order: Experimental value, sum of powers in rate law, can be zero, fractional, or negative.
  Molecularity: Theoretical value, number of colliding molecules in an elementary reaction, can only be a non-zero positive integer ($1, 2, 3$1, 2, 3).

3.2 Inorganic Chemistry 2-Markers

- **Q20 [d-Block] Why is $E^\circ_{Cu^{2+}/Cu}$ ECu2+/Cu∘ positive ($+0.34\ \text{V}$+0.34 V)?**
  Answer: The high enthalpy of atomization and high ionization energies ($\text{IE}_1 + \text{IE}_2$ IE1 + IE2) of Copper are not balanced by its hydration enthalpy.
- **Q21 [d-Block] Why are $\text{Zn}, \text{Cd}, \text{Hg}$Zn, Cd, and Hg not considered typical transition metals?**
  Answer: Because they have fully filled dd-orbitals ($(n-1)d^{10}$(n − 1)d10) both in their ground atomic state and in their common oxidation states ($Zn^{2+}, Cd^{2+}, Hg^{2+}$ Zn2+, Cd2+, Hg2+).
- **Q22 [Coordination] What are Ambidentate Ligands? Give two examples.**
  Answer: Unidentate ligands that can coordinate through either of two different donor atoms (e.g., NO_2^- NO2− coordinating via NN [nitro] or OO [nitrito]; SCN^- SCN− coordinating via SS [thiocyanato] or NN [isothiocyanato]).

3.3 Organic Chemistry 2-Markers

- **Q23 [Haloarenes] Why are haloarenes extremely less reactive towards nucleophilic substitution than haloalkanes?**
  Answer: (1) **Resonance effect**: C—Cl bond acquires partial double-bond character. (2) **Hybridization**: Carbon is sp^2 sp2 hybridized (more electronegative, holds bonding electrons tighter than sp^3 sp3).
- **Q24 [Phenols] Why is ortho-nitrophenol steam volatile, while para-nitrophenol is not?**
  Answer: Ortho-nitrophenol has **intramolecular H-bonding** (lower boiling point). Para-nitrophenol has **intermolecular H-bonding**, causing molecular association and higher boiling point.
- **Q25 [Amines] Explain the Hinsberg Test to distinguish $1^\circ, 2^\circ,$ 1∘, 2∘, and 3° 3∘ amines.**
  - 1° 1∘ Amine: Forms precipitate **soluble in** KOH**KOH**.
  - 2° 2∘ Amine: Forms precipitate **insoluble in** KOH**KOH**.
  - 3° 3∘ Amine: **Does not react**.
- **Q26 [Biomolecules] Give two differences between DNA and RNA.**
  1. DNA contains **2-deoxyribose** sugar and **Thymine (T)**; RNA contains **ribose** sugar and **Uracil (U)**.
  2. DNA is double-stranded; RNA is single-stranded.

PART 4: THE 1-MARK MCQS & OBJECTIVES (SECTION A — 10 MARKS)

1. **What is the SI unit of rate constant for a zero-order reaction?**
   → → $\mathrm{mol \cdot L^{-1} \cdot s^{-1}}$ **mol · L−1 · s−1**
2. **Which of the following ions is diamagnetic and colorless?**
   → → $\mathrm{Sc^{3+}}$ **Sc3+** (or $\mathrm{Zn^{2+}}$ Zn2+, d^0/d^{10} d0/d10)
3. **An azeotropic mixture boils at a lower temperature than either component when it exhibits:**
   → → **Positive deviation from Raoult's Law**
4. **Coordination number of** Co**Co in** $[\mathrm{Co(en)_3}]^{3+}$ **[Co(en)3]3+ is:**
   → → 6**6** *(Ethylenediamine is bidentate:* $3 \times 2 = 6$*3 × 2 = 6)*
5. **What is Lucas Reagent?**
   → → **Anhydrous** $\mathrm{ZnCl_2}$ **ZnCl2 + concentrated** HCl**HCl**

6. **The reduction of an acyl chloride to an aldehyde using Pd-BaSO_4 Pd-BaSO4 is known as:**
   → → **Rosenmund Reduction**
7. **Which vitamin is water-soluble and must be regularly taken in the diet?**
   → → **Vitamin C** (or Vitamin B-complex)
8. **In an alkaline medium, KMnO_4 KMnO4 is reduced to:**
   → → MnO_2 **MnO2** (Oxidation state changes from $+7$+7 to $+4$+4)
9. **Carbylamine test is given only by:**
   → → **Primary (1° 1∘) aliphatic and aromatic amines**
10. **The quantity of electricity required to deposit $1\ \text{mole}$1 mole of AlAl from Al^{3+} Al3+ is:**
    → → $3\ \text{Faraday}$**3 Faraday (**$3 \times 96500\ \text{C}$**3 × 96500 C)**